

Estela Pérez Luque, PhD

Ergonomics Specialist Engineer (Occupant Packaging)

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PROFILE

Automotive Ergonomics Engineer specializing in occupant packaging and DMU-based ergonomic validation for vehicle interior development. I evaluate and optimize vehicle architecture concepts using real-vehicle driver data, digital human modelling, and optimization methods to support clearance, reachability, visibility, comfort, and usability targets. I collaborate across engineering and research partners to translate ergonomic evidence into decision-ready recommendations in early development phases. Motivated to contribute to Mercedes-Benz vehicle architecture by applying rigorous ergonomic validation to current and future vehicle programs.

TECHNICAL SKILLS

Simulation & Occupant Packaging: occupant packaging analysis, ergonomic evaluation methods, anthropometric variability assessment, DMU, posture & motion prediction, simulation-based multi-objective optimization (SBMOO), CAD-integrated ergonomics workflows. Digital Human Modelling: IPS IMMA, Siemens Jack, Anybody; CAD tools: Autodesk Inventor, SolidWorks, NX, Teamcenter.

Experimental Design and Human Data: Study design & protocol definition, ethical approval preparation, real-vehicle & lab-based data collection, motion capture (MOVE4D, XSENS, optical systems), sensor synchronization, anthropometric data processing, data cleaning, validation workflows, reproducible analysis pipelines.

Tools & Modelling: MATLAB, Python, predictive modelling, optimization-based modelling, data-driven modelling, statistical regression, inverse kinematics, time-series analysis, data visualization, structured processing pipelines.

Languages: English (C1), Spanish (Native), Swedish (B1).

PROFESSIONAL EXPERIENCE

Ergonomics Researcher, University of Skövde (Sweden)

Sep 2020 – Oct 2025

- **Ergonomic Concept Development and Validation:** I led validation of occupant packaging concepts for real vehicle configurations, evaluating H-point, eye-point, and steering wheel point positioning by digital mock-ups, considering different aspects (comfort, visibility, reachability, etc.), resulting in more accurate, inclusive models and optimized processes that enhance ergonomic decision support in early vehicle development.
- **Cross-functional Collaboration and User Needs:** I directed research studies with Volvo Cars, Volvo Trucks, Scania, Texas Tech University, and the University of Antwerp by aligning objectives, timelines, and expectations across both technical and non-technical teams, translating complex analytical results into actionable insights relevant for automotive ergonomics and safety-focused design, which resulted in more than 15 peer-reviewed publications.

Visiting Researcher - University of Antwerp (Belgium)

Sep 2024 – Mar 2025

- **Cross-Functional Coordination, Study Governance, and Data Pipelines:** I oversaw a six-month international research stay and managed the technical preparations across institutions. I took full responsibility for ethics applications and experimental protocol design for motion-capture, and I developed structured, reproducible data pipelines for advanced 4D body-scanning and motion-capture data. I provided reliable workflows and documentation that enabled multi-site collaborators to conduct consistent, data-driven validation and generate high-quality outputs.

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Co-founder and R&D Lead - Swe-Ex AB (Startup) (Sweden)

Sep 2021 – Mar 2023

- **Product Development and Prototype Validation:** I co-founded a wearable exoskeleton startup and led concept development and prototyping with an emphasis on user safety and ergonomics. I translated user needs into measurable design goals within real-world constraints and managed the iteration cycle from CAD to physical prototypes using Autodesk Inventor and SolidWorks. This resulted in three functional prototype versions with 3D-printed, textile, and steel parts. I verified core functions through prototype testing and stakeholder demos, demonstrating full ownership from concept to physical validation.

Research Assistant - University of Skövde (Sweden)

Jun 2019 – Sep 2020

- **Product demonstration and Communication:** I performed field-based ergonomic assessments (RULA, REBA) and workflow analyses for 6+ industry partners, pinpointing high-risk postures and suggesting layout changes to enhance usability and safety. I assisted partners in applying ergonomic insights during product development discussions and turning findings into practical deliverables.

EDUCATION

PhD in Informatics, University of Skövde (Sweden)

Sep 2020 – Oct 2025

[PhD Thesis](#): Human Posture and Motion Prediction for Automotive Ergonomics Design.

Focused on validation of posture and motion prediction methods for occupant packaging and vehicle interior evaluation using motion capture data and digital human modelling (DMU-based workflows).

Emphasized simulation accuracy, anthropometric variability, and integration of ergonomic evaluation into early vehicle development phases.

Master of Science in Intelligent Automation, University of Skövde (Sweden)

Aug 2018 – Jun 2019

Bachelor of Science in Mechanical Engineering, University of Skövde (Sweden)

Aug 2017 – Jun 2018

Degree in Mechanical Engineering, University of Málaga (Spain)

Aug 2012 – Jun 2018

RESEARCH PUBLICATIONS

Full list available at: [Google Scholar – Estela Pérez Luque](#)